

For the Use Only of a Registered Medical Practitioner

PRESCRIBING INFORMATION

XOLSTAT TABLETS (Rosuvastatin Calcium Tablets 5 mg / 10 mg / 20 mg)

COMPOSITION

XOLSTAT 5 mg

Each film coated tablet contains
Rosuvastatin Calcium equivalent to
Rosuvastatin 5 mg

XOLSTAT 10 mg

Each film coated tablet contains
Rosuvastatin Calcium equivalent to
Rosuvastatin 10 mg

XOLSTAT 20 mg

Each film coated tablet contains
Rosuvastatin Calcium equivalent to
Rosuvastatin 20 mg

Excipients:

XOLSTAT 5 mg tablets also contain: Lactose monohydrate, crospovidone, microcrystalline cellulose, sodium citrate, magnesium stearate, opadry yellow (hypromellose, titanium dioxide, polyethylene glycol / macrogol, yellow iron oxide).

XOLSTAT 10 mg & 20 mg tablets also contain: Lactose monohydrate, crospovidone, microcrystalline cellulose, sodium citrate, magnesium stearate, opadry pink (hypromellose, titanium dioxide, polyethylene glycol / macrogol, red iron oxide).

PRODUCT DESCRIPTION

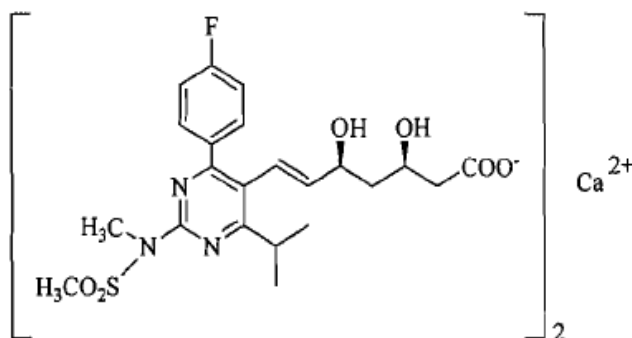
XOLSTAT 5 mg: Light yellow to yellow colored round film coated tablets with 'RT1' on one side and plain on the other side.

XOLSTAT 10 mg: Light pink to pink colored round film coated tablets with 'RT2' on one side and plain on the other side.

XOLSTAT 20 mg: Light pink to pink colored round film coated tablets with 'RT3' on one side and plain on the other side.

DESCRIPTION

XOLSTAT tablets contain rosuvastatin calcium, which is a synthetic lipid-lowering agent for oral administration. The empirical formula for rosuvastatin calcium is $(C_{22}H_{27}FN_3O_6S)_2Ca$ and the molecular weight is 1001.14. The chemical name for rosuvastatin calcium is bis[(*E*)-7-[4-(4-fluorophenyl)-6-isopropyl-2-[methyl(methylsulfonyl)amino] pyrimidin-5-yl](3*R*,5*S*)-3,5-dihydroxyhept-6-enoic acid] calcium salt with the following structural formula:



ROSUVASTATIN CALCIUM

INDICATIONS

Rosuvastatin is indicated as an adjunct to diet, at least equivalent to the Adult Treatment Panel III (ATP III TLC diet), for the reduction of elevated total cholesterol, LDL-cholesterol, ApoB, the total cholesterol: HDL-cholesterol ratio and triglycerides and for increasing HDL-C, in hyperlipedemic and dyslipidemic conditions, when response to diet and exercise alone has been inadequate including:

Prevention of Cardiovascular Events. In adult patients with an increased risk of atherosclerotic cardiovascular disease based on the presence of cardiovascular disease risk markers such as an elevated hsCRP level, age, hypertension, low HDL-C, smoking or a family history of premature coronary heart disease, rosuvastatin is indicated to reduce total mortality and the risk of major cardiovascular events (cardiovascular death, stroke, MI, unstable angina, or arterial revascularization).

Rosuvastatin is indicated as an adjunct to diet for the treatment of patients with primary dysbetalipoproteinemia (Type III hyperlipoproteinemia).

Primary hypercholesterolemia (Type IIa excluding heterozygous familial hypercholesterolemia); and severe non-familial hypercholesterolemia).

Combined (mixed) dyslipidemia (Type IIb).

Homozygous familial hypercholesterolemia where rosuvastatin is used either alone or as an adjunct to diet and other lipid lowering treatment such as apheresis.

Rosuvastatin is indicated as adjunctive therapy to diet to slow the progression of atherosclerosis in adult patients as part of a treatment strategy to lower Total-C and LDL-C to target levels.

DOSE AND METHOD OF ADMINISTRATION

Patients should be placed on a standard cholesterol-lowering diet (at least equivalent to the Adult Treatment Panel III (ATP III TLC diet)) before receiving rosuvastatin, and should continue on this diet during treatment with rosuvastatin. If appropriate, a program of weight control and physical exercise should be implemented.

Prior to initiating therapy with rosuvastatin, secondary causes for elevations in plasma lipid levels should be excluded. A lipid profile should also be performed. After initiation or upon titration of rosuvastatin, lipid levels should be analyzed within 2-4 weeks and the dosage adjusted accordingly.

The usual recommended starting dose of rosuvastatin is 10 mg once daily. However, initiation of therapy with 5 mg once daily should be considered for special patient populations or patients requiring less aggressive LDL-C reductions. The choice of starting dose should take into account the individual patients' cholesterol level and future cardiovascular risk as well as the potential risk for adverse reactions. Rosuvastatin may be taken in the morning or evening with or without food. The majority of patients are controlled at the 10 mg dose. However, if necessary, dose adjustments to the next dose level can be made after 4-week intervals. The maximum response is usually achieved within 2-4 weeks and is maintained during chronic therapy. Increasing the dose to 40 mg should be reserved for patients with severe hypercholesterolemia excluding heterozygous familial hypercholesterolemia at high cardiovascular risk (in particular dose with familial hypercholesterolemia), who do not achieve their treatment goal on 20 mg and should only be initiated under specialist supervision. The physician who elects to use rosuvastatin at a dose higher than 20 mg should periodically re-evaluate the long term risk/benefit of rosuvastatin for the individual patient. Rosuvastatin should be prescribed with caution in patients with pre-disposing factors for myopathy/rhabdomyolysis.

The dosage of rosuvastatin should be individualized according to baseline LDL-C, total-C/HDL-C ratio and/or TG levels, the recommended target lipid values and the patient response.

Lipid levels should be monitored periodically and, if necessary, the dose of rosuvastatin adjusted based on target lipid levels recommended by guidelines.

Dosage in patients with renal insufficiency

The usual dose range applies in patients with mild to moderate renal impairment.

The use of rosuvastatin in patients with severe renal impairment is contraindicated.

Dosage in patients with hepatic insufficiency

There was no increase in systemic exposure to rosuvastatin in subjects with Child-Pugh scores of 7 or below. However, increased systemic exposure has been reported in subjects with Child-pugh scores of 8 and 9. In these patients an assessment of renal function should be considered. There is no experience in subjects with Child-Pugh scores above 9. Rosuvastatin is contraindicated in patients with active liver disease.

Use in the elderly

The overall frequency of adverse events and types of adverse events have been reported to be similar in patients above and below 65 years of age. The efficacy of rosuvastatin in the geriatric population (≥ 65 years of age) has been reported to be comparable to the efficacy reported in the non-elderly.

Use in Children below 10 years

The safety and effectiveness in children have not been established. In children and adolescents with homozygous familial hypercholesterolemia experience is limited to eight patients (aged 8 years and above).

Dosage on Asian Patients

Initiation of rosuvastatin therapy with 5 mg once daily should be considered for Asian patients. The potential for increased systemic exposures relative to Caucasians is relevant when considering escalation of dose in cases where hypercholesterolemia excluding heterozygous familial hypercholesterolemia is not adequately controlled at doses of 5, 10 or 20 mg once daily.

Concomitant therapy

The effect of rosuvastatin on LDL-C and total-C may be enhanced when used in combination with a bile acid binding resin. If rosuvastatin is used in combination with gemfibrozil, the dose of rosuvastatin should be limited to 10 mg once daily.

Interactions requiring dose adjustments

Gemfibrozil: Increased systemic exposure to rosuvastatin has been reported in subjects taking concomitant rosuvastatin and gemfibrozil. Patients taking this combination should not exceed a dose of rosuvastatin 10 mg once daily (see WARNINGS AND PRECAUTIONS AND DRUG INTERACTIONS).

Dosage in patients with pre-disposing factors to myopathy

The recommended start dose is 5 mg in patients with predisposing factors to myopathy.

Concomitant therapy

Rosuvastatin is a substrate of various transporter proteins (e.g. OATP1B1 and BCRP). The risk of myopathy (including rhabdomyolysis) is increased when rosuvastatin is administered concomitantly with certain medicinal

products that may increase the plasma concentration of rosuvastatin due to interactions with these transporter proteins (e.g. certain protease inhibitors including combinations of ritonavir with atazanavir, lopinavir, and/or tipranavir). Whenever possible, alternative medications should be considered, and, if necessary, consider temporarily discontinuing rosuvastatin therapy. In situations where co-administration of these medicinal products with rosuvastatin is unavoidable, the benefit and the risk of concurrent treatment and rosuvastatin dosing adjustments should be carefully considered.

CONTRAINDICATIONS

XOLSTAT (rosuvastatin) tablets are contraindicated in:

- patients with hypersensitivity to rosuvastatin or to any of the excipients of the product. Hypersensitivity reactions including rash, pruritus, urticaria, and angioedema have been reported with rosuvastatin (see UNDESIRABLE EFFECTS).
- patients with active liver disease including unexplained, persistent elevations of serum transaminases and any serum transaminase elevation exceeding 3 times the upper limit of normal (ULN).
- patients with myopathy.
- patients receiving concomitant cyclosporine.
- during pregnancy, while breast-feeding and in women of childbearing potential not using appropriate contraceptive measures.

WARNINGS AND PRECAUTIONS

Renal effects

Proteinuria, detected by dipstick testing and mostly tubular in origin and microscopic hematuria have been reported in patients treated with higher doses of rosuvastatin, in particular 40 mg, where it was transient or intermittent in most cases. Proteinuria has not been reported to be predictive of acute or progressive renal disease. An assessment of renal function should be considered during routine follow-up of patients treated with a dose of 40 mg (see UNDESIRABLE EFFECTS).

Renal impairment

Rosuvastatin exposure is not influenced by mild to moderate renal impairment ($CL_{Cr} \geq 30$ mL/min/1.73 m²). Exposure to rosuvastatin is increased to a clinically significant extent in patients with severe renal impairment ($CL_{Cr} < 30$ mL/min/1.73 m²) who are not receiving hemodialysis and dose adjustment is required.

Skeletal muscle effects

Gemfibrozil increases the risk of myopathy when given concomitantly with some HMG-CoA reductase inhibitors. Therefore, the combination of rosuvastatin and gemfibrozil is not recommended. The benefit of further alterations in lipid levels by the combined use of rosuvastatin with fibrates or niacin should be carefully weighed against the potential risks of such combinations.

Effects on skeletal muscle e.g. myalgia, myopathy and, rarely, rhabdomyolysis have been reported in rosuvastatin-treated patients with all doses and in particular with doses > 20 mg. Very rare cases of rhabdomyolysis have been reported with the use of ezetimibe in combination with HMG-CoA reductase inhibitors. A pharmacodynamic interaction cannot be excluded (see DRUG INTERACTIONS) and caution should be exercised with their combined use.

Rosuvastatin should be prescribed with caution in patients with predisposing factors for myopathy (e.g., age $\geq$ 65 years, inadequately treated hypothyroidism, renal impairment).

Creatine kinase measurement

Creatine Kinase (CK) should not be measured following strenuous exercise or in the presence of a plausible alternative cause of CK increase which may confound interpretation of the result. If CK levels are significantly elevated at baseline ($>5\times\text{ULN}$) a confirmatory test should be carried out within 5 to 7 days. If the repeat test confirms a baseline CK $>5\times\text{ULN}$, treatment should not be started.

Before treatment

Rosuvastatin, as with other HMG-CoA reductase inhibitors, should be prescribed with caution in patients with predisposing factors for myopathy/rhabdomyolysis. Such factors include:

- renal impairment
- hypothyroidism
- personal or family history of hereditary muscular disorders
- previous history of muscular toxicity with another HMG-CoA reductase inhibitor or fibrate
- alcohol abuse
- age >70 years
- situations where an increase in plasma levels may occur (see PHARMACODYNAMIC AND PHARMACOKINETIC PROPERTIES).
- concomitant use of fibrates.

In such patients the risk of treatment should be considered in relation to possible benefit and clinical monitoring is recommended. If CK levels are significantly elevated at baseline ($>5\times\text{ULN}$), treatment should not be started.

Whilst on treatment

All patients should be advised to promptly report to their physician inexplicable/unexplained muscle pain, tenderness, cramps or weakness, particularly if accompanied by malaise or fever or if muscle signs and symptoms persist after discontinuing rosuvastatin.

CK levels should be measured in these patients. Therapy should be discontinued if CK levels are markedly elevated ($>5\times\text{ULN}$) or if muscular symptoms are severe and cause daily discomfort (even if CK levels are $\leq 5\times\text{ULN}$). If symptoms resolve and CK levels return to normal, then consideration should be given to re-introducing

rosuvastatin or an alternative HMG-CoA reductase inhibitor at the lowest dose with close monitoring. Routine monitoring of CK levels in asymptomatic patients is not warranted.

The risk of myopathy during treatment with rosuvastatin may be increased with concurrent administration of some other lipid-lowering therapies (fibrates or niacin), gemfibrozil, cyclosporine, lopinavir/ritonavir, or atazanavir/ritonavir.

An increase in the incidence of myositis and myopathy has been reported in patients receiving other HMG-CoA reductase inhibitors together with fibric acid derivatives including gemfibrozil, cyclosporine, nicotinic acid, azole antifungals, protease inhibitors and macrolide antibiotics. Gemfibrozil increases the risk of myopathy when given concomitantly with some HMG-CoA reductase inhibitors. Therefore, the combination of rosuvastatin and gemfibrozil is not recommended. The benefit of further alterations in lipid levels by the combined use of rosuvastatin with fibrates or niacin should be carefully weighed against the potential risks of such combinations. (see DRUG INTERACTIONS; DOSAGE AND ADMINISTRATION; UNDESIRABLE EFFECTS).

Rosuvastatin should not be used in any patient with an acute, serious condition suggestive of myopathy or predisposing to the development of renal failure secondary to rhabdomyolysis (e.g. sepsis, hypotension, dehydration, major surgery, trauma, severe metabolic, endocrine and electrolyte disorders; or uncontrolled seizures).

There have been very rare reports of an immune-mediated necrotizing myopathy (IMNM) during or after treatment with some statin. IMNM is clinically characterized by:

- persistent proximal muscle weakness and elevated serum creatine kinase, which persist despite discontinuation of statin treatment;
- muscle biopsy showing necrotizing myopathy without significant inflammation
- improvement with immunosuppressive agents.

Liver effects

As with other HMG-CoA reductase inhibitors, rosuvastatin should be used with caution in patients who consume excessive quantities of alcohol and/or have a history of liver disease (including chronic liver disease).

It is recommended that liver function tests be carried out prior to treatment initiation and if signs or symptoms of liver injury occur, and 3 months following the initiation of treatment. Rosuvastatin should be discontinued or the dose reduced if the level of serum transaminases is greater than 3 times the upper limit of normal. The reporting rate for serious hepatic events (consisting mainly of increased hepatic transaminases) in post-marketing use is higher at the 40 mg dose.

In patients with secondary hypercholesterolemia excluding heterozygous familial hypercholesterolemia caused by hypothyroidism or nephrotic syndrome, the underlying disease should be treated prior to initiating therapy with rosuvastatin.

Race

An increase in exposure has been reported in Asian subjects compared with Caucasians (see DOSE AND METHOD OF ADMINISTRATION; CONTRAINDICATIONS; PHARMACODYNAMIC AND PHARMACOKINETIC PROPERTIES).

Protease inhibitors

Increased systemic exposure to rosuvastatin has been reported in subjects receiving rosuvastatin concomitantly with various protease inhibitors in combination with ritonavir. Consideration should be given both to the benefit of lipid lowering by use of rosuvastatin in HIV patients receiving protease inhibitors and the potential for increased rosuvastatin plasma concentrations when initiating and up titrating rosuvastatin doses in patients treated with protease inhibitors. The concomitant use with certain protease inhibitors is not recommended unless the dose of rosuvastatin is adjusted.

Interstitial lung disease

Exceptional cases of interstitial lung disease have been reported with some statins, especially with long term therapy (see UNDESIRABLE EFFECTS). Presenting features can include dyspnoea, non-productive cough and deterioration in general health (fatigue, weight loss and fever). If it is suspected a patient has developed interstitial lung disease, statin therapy should be discontinued.

Diabetes mellitus

Some evidence suggests that statins (including rosuvastatin) as a class raise blood fasting serum glucose levels and HbA1c, and in some patients at high risk of future diabetes, may produce a level of hyperglycemia where formal diabetes care is appropriate. This risk, however, is outweighed by the reduction in vascular risk with statins and therefore should not be a reason for stopping statin treatment. Patients at risk (fasting glucose 5.6 to 6.9 mmol/L, BMI >30 kg/m², raised triglycerides, hypertension) should be monitored both clinically and biochemically according to national guidelines.

Concomitant coumarin anticoagulants

Caution should be exercised when anticoagulants are given in conjunction with rosuvastatin because of its potentiation of the effect of coumarin-type anticoagulants in prolonging the prothrombin time/INR. In patients taking coumarin anticoagulants and rosuvastatin concomitantly, INR should be determined before starting rosuvastatin and frequently enough during early therapy to ensure that no significant alteration of INR occurs (see DRUG INTERACTIONS).

Pediatric population

The evaluation of linear growth (height), weight, BMI (body mass index), and secondary characteristics of sexual maturation by Tanner staging in pediatric patients 10 to 17 years of age taking rosuvastatin is limited to a one-year period.

Lactose intolerance

This product contains lactose as an excipient. Therefore, patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine.

Effects on ability to drive and use machines

Pharmacology testing revealed no evidence of a sedative effect of rosuvastatin. From the safety profile, rosuvastatin is not expected to adversely affect the ability to drive or use machines.

USE IN SPECIAL POPULATIONS

The safety of rosuvastatin during pregnancy and whilst breast-feeding has not been established. Women of child-bearing potential should use appropriate contraceptive measures (See CONTRAINDICATIONS).

DRUG INTERACTIONS

Effect of co-administered medicinal products on rosuvastatin

Transporter protein inhibitors: Rosuvastatin is a substrate for certain transporter proteins including the hepatic uptake transporter OATP1B1 and efflux transporter BCRP. Concomitant administration of rosuvastatin with medicinal products that are inhibitors of these transporter proteins may result in increased rosuvastatin plasma concentrations and an increased risk of myopathy.

Ciclosporin: Ciclosporin increased rosuvastatin exposure and may result in increased risk of myopathy. Therefore, in patients taking ciclosporin, the dose of rosuvastatin should not exceed 5 mg once daily.

Protease inhibitors: Although the exact mechanism of interaction is unknown, concomitant protease inhibitor use may strongly increase rosuvastatin exposure (see Table below). For instance, co-administration of 10 mg rosuvastatin and a combination product of two protease inhibitors (300 mg atazanavir / 100 mg ritonavir) in healthy volunteers was associated with an approximately three-fold and seven-fold increase in rosuvastatin AUC and C_{max} respectively. The concomitant use of rosuvastatin and some protease inhibitor combinations may be considered after careful consideration of rosuvastatin dose adjustments based on the expected increase in rosuvastatin exposure.

Gemfibrozil and other lipid-lowering products: Concomitant use of rosuvastatin and gemfibrozil has been reported to result in a 2-fold increase in rosuvastatin C_{max} and AUC. Basis reported data, no pharmacokinetic relevant interaction with fenofibrate is expected, however a pharmacodynamic interaction may occur. Gemfibrozil, fenofibrate, other fibrates and lipid lowering doses (> or equal to 1g/day) of niacin (nicotinic acid) increase the risk of myopathy when given concomitantly with HMG-CoA reductase inhibitors, probably because they can produce myopathy when given alone. These patients should also start with the 5 mg dose.

Ezetimibe: Concomitant use of 10 mg rosuvastatin and 10 mg ezetimibe has been reported to result in a 1.2 fold increase in AUC of rosuvastatin in hypercholesterolaemic subjects. A pharmacodynamic interaction, in terms of adverse effects, between rosuvastatin and ezetimibe cannot be ruled out.

Antacid: The simultaneous dosing of rosuvastatin with an antacid suspension containing aluminium and magnesium hydroxide has been reported to result in a decrease in rosuvastatin plasma concentration of approximately 50%. This effect was mitigated when the antacid was dosed 2 hours after rosuvastatin. The clinical relevance of this interaction has not been reported.

Fusidic Acid: Interaction with rosuvastatin and fusidic acid have not been reported. As with other statins, muscle related events, including rhabdomyolysis, have been reported in post-marketing experience with rosuvastatin and fusidic acid given concurrently. Patients should be closely monitored and temporary suspension of rosuvastatin treatment may be appropriate.

Erythromycin: Concomitant use of rosuvastatin and erythromycin has been reported to result in a 20% decrease in AUC (0-t) and a 30% decrease in C_{max} of rosuvastatin. This interaction may be caused by the increase in gut motility caused by erythromycin.

Cytochrome P450 enzymes: Rosuvastatin is neither an inhibitor nor an inducer of cytochrome P450 isoenzymes. In addition, rosuvastatin is a poor substrate for these isoenzymes. Therefore, drug interactions resulting from cytochrome P450-mediated metabolism are not expected. No clinically relevant interactions have been reported between rosuvastatin and either fluconazole (an inhibitor of CYP2C9 and CYP3A4) or ketoconazole (an inhibitor of CYP2A6 and CYP3A4).

Interactions requiring rosuvastatin dose adjustments (see also Table below)

When it is necessary to co-administer rosuvastatin with other medicinal products known to increase exposure to rosuvastatin, doses of rosuvastatin should be adjusted. It is recommended that prescribers consult the relevant product information when considering administration of such products together with rosuvastatin.

If medicinal product is observed to increase rosuvastatin AUC approximately 2-fold or higher, the starting dose of rosuvastatin should not exceed 5 mg once daily. The maximum daily dose of rosuvastatin should be adjusted so that the expected rosuvastatin exposure would not likely exceed that of a 40 mg daily dose of rosuvastatin taken without interacting medicinal products, for example a 5 mg dose of rosuvastatin with ciclosporin (7.1-fold increase in exposure), a 10 mg dose of rosuvastatin with ritonavir/atazanavir combination (3.1-fold increase) and a 20 mg dose of rosuvastatin with gemfibrozil (1.9-fold increase).

If medicinal product is observed to increase rosuvastatin AUC less than 2-fold, the starting dose need not be decreased but caution should be taken if increasing the rosuvastatin dose above 20mg.

Protease Inhibitors: Coadministration of rosuvastatin with certain protease inhibitors or combination of protease inhibitors may increase the rosuvastatin exposure, (AUC) up to 7-fold. Dose adjustment are needed depending on the level of effect on rosuvastatin exposure

Table: Effect of co-administered medicinal products on rosuvastatin exposure (AUC; in order of decreasing magnitude)

2-fold or greater than 2-fold increase in AUC of rosuvastatin		
Interacting drug dose regimen	Rosuvastatin dose regimen	Change in rosuvastatin AUC*
Sofosbuvir/velpatasvir/voxilaprevir (400 mg-100 mg-100 mg) + Voxilaprevir (100 mg) once daily for 15 days	10mg single dose	7.39 -fold ↑
Ciclosporin 75 mg BID to 200 mg BID, 6 months	10 mg OD, 10 days	7.1-fold ↑
Darolutamide 600 mg BID, 5 days	5mg, single dose	5.2-fold ↑
Regorafenib 160 mg OD, 14 days	5mg, single dose	3.8 -fold ↑
Atazanavir 300 mg/ritonavir 100 mg OD, 8 days	10 mg, single dose	3.1-fold ↑
Simeprevir 150 mg OD, 7 days	10 mg, single dose	2.8-fold ↑
Velpatasvir 100 mg OD	10 mg single dose	2.69 -fold ↑
Ombitasvir 25 mg/paritaprevir 150 mg/ritonavir 100 mg/dasabuvir 400 mg BID	5mg single dose	2.59-fold ↑
Grazoprevir 200 mg/elbasvir 50 mg OD	10mg single dose	2.26-fold ↑
Glecaprevir 400 mg/pibrentasvir 120 mg OD for 7 days	5mg once daily	2.2-fold ↑
Lopinavir 400 mg/ritonavir 100 mg BID, 17 days	20 mg OD, 7 days	2.1-fold ↑
Clopidogrel 300 mg loading, followed by 75 mg at 24 hours ³	20 mg, single dose ³	2-fold ↑
Gemfibrozil 600 mg BID, 7 days	80 mg, single dose	1.9-fold ↑
Less than 2-fold increase in AUC of rosuvastatin		
Interacting drug dose regimen	Rosuvastatin dose regimen	Change in rosuvastatin AUC*
Eltrombopag 75 mg OD, 5 days	10 mg, single dose	1.6-fold ↑
Darunavir 600 mg/ritonavir 100 mg BID, 7 days	10 mg OD, 7 days	1.5-fold ↑
Tipranavir 500 mg/ritonavir 200 mg BID, 11 days	10 mg, single dose	1.4-fold ↑
Dronedarone 400 mg BID	Not available	1.4-fold ↑
Itraconazole 200 mg OD, 5 days	10 mg, single dose	**1.4-fold ↑
Ezetimibe 10 mg OD, 14 days	10 mg, OD, 14 days	**1.2-fold ↑
Decrease in AUC of rosuvastatin		
Interacting drug dose regimen	Rosuvastatin dose regimen	Change in rosuvastatin AUC*
Erythromycin 500 mg QID, 7 days	80 mg, single dose	20% ↓
Baicalin 50 mg TID, 14 days	20 mg, single dose	47% ↓
*Data given as x-fold change represent a simple ratio between co-administration and rosuvastatin alone. Data given as % change represent % difference relative to rosuvastatin alone. Increase is indicated as "↑", decrease as "↓". **Several interaction data have been reported at different rosuvastatin dosages, the table shows the most significant ratio AUC = area under curve; OD = once daily; BID = twice daily; TID = three times daily; QID = four times daily		

The following medicinal product/combinations did not have a clinically significant effect on the AUC ratio of rosuvastatin at coadministration:

Aleglitazar 0.3 mg 7 days dosing; Fenofibrate 67 mg 7 days TID dosing; Fluconazole 200mg 11 days OD dosing; Fosamprenavir 700 mg/ritonavir 100 mg 8 days BID dosing; Ketoconazole 200 mg 7 days BID dosing; Rifampin 450 mg 7 days OD dosing; Silymarin 140 mg 5 days TID dosing.

Other medications: Concurrent use of fibrates may cause severe myositis and myoglobinuria.

Effect of rosuvastatin on co-administered medicinal products

Vitamin K antagonists: As with other HMG-CoA reductase inhibitors, the initiation of treatment or dosage up-titration of rosuvastatin in patients treated concomitantly with vitamin K antagonists (e.g. warfarin or another coumarin anticoagulant) may result in an increase in International Normalised Ratio (INR). Discontinuation or down-titration of rosuvastatin may result in a decrease in INR. In such situations, appropriate monitoring of INR is desirable.

Oral contraceptive/hormone replacement therapy (HRT): Concomitant use of rosuvastatin and an oral contraceptive have been reported to result in an increase in ethinyl estradiol and norgestrel AUC of 26% and 34%, respectively. These increased plasma levels should be considered when selecting oral contraceptive doses. There are no pharmacokinetic data reported in subjects taking concomitant rosuvastatin and HRT and therefore a similar effect cannot be excluded. However, the combination has been reported to be extensively used in women and was well tolerated.

Other medicinal products: Based on reported data from specific interaction no clinically relevant interaction with digoxin is expected.

Paediatric population: Interaction data have only been reported in adults. The extent of interactions in the paediatric population is not known.

UNDESIRABLE EFFECTS

There have been rare post-marketing reports of cognitive impairment (e.g. memory loss, forgetfulness, amnesia, memory impairment, confusion) associated with statin use. These issues have been reported for all statins. The reports are generally non-serious and reversible upon statin discontinuation, with variable times to symptom onset (1 day to years) and symptom resolution (median 3 weeks).

Increases in HbA1c and fasting blood glucose have been reported with statins. The risk of hyperglycemia, however, is outweighed by the reduction in vascular risk with statins.

The adverse reactions reported with rosuvastatin are generally mild and transient.

Tabulated list of adverse reactions

The following table presents the adverse reaction profile for rosuvastatin. Adverse reactions listed below are classified according to frequency and system organ class (SOC).

The frequencies of adverse reactions are ranked according to the following convention: Common ($\geq 1/100$ to $< 1/10$); Uncommon ($\geq 1/1,000$ to $< 1/100$); Rare ($\geq 1/10,000$ to $< 1/1,000$); Very rare ($< 1/10,000$); Not known (cannot be estimated from the available data).

Table: Adverse reactions reported with rosuvastatin

System organ class	Common	Uncommon	Rare	Very rare	Not known
Blood and lymphatic system disorders			Thrombocytopenia		
Immune system disorders			Hypersensitivity reactions including angioedema		
Endocrine disorders	Diabetes mellitus ¹				
Psychiatric disorders					Depression
Nervous system disorders	Headache Dizziness			Polyneuropathy Memory loss	Sleep disturbances (including insomnia and nightmares) Peripheral neuropathy
Respiratory, thoracic and mediastinal disorders					Cough Dyspnoea
Gastro-intestinal disorders	Constipation Nausea Abdominal pain		Pancreatitis		Diarrhoea
Hepatobiliary disorders			Increased hepatic transaminases	Jaundice Hepatitis	
Skin and subcutaneous tissue disorders		Pruritis Rash Urticaria			Stevens-Johnson syndrome
Musculo-skeletal and connective tissue disorders	Myalgia		Myopathy (including myositis) Rhabdomyolysis	Arthralgia	Immune-mediated necrotising myopathy
Renal and urinary				Haematuria	

System organ class	Common	Uncommon	Rare	Very rare	Not known
disorders					
Reproductive system and breast disorders				Gynaecomastia	
General disorders and administration site conditions	Asthenia				Oedema
¹ Frequency will depend on the presence or absence of risk factors (fasting blood glucose $\geq$ 5.6 mmol/L, BMI $>$ 30 kg/m ² , raised triglycerides, history of hypertension).					

As with other HMG-CoA reductase inhibitors, the incidence of adverse drug reactions tends to be dose dependent.

Renal effects: Proteinuria, detected by dipstick testing and mostly tubular in origin, has been reported in patients treated with rosuvastatin. Shifts in urine protein from none or trace to ++ or more have been reported in $<$ 1% of patients at some time during treatment with 10 and 20 mg, and in approximately 3% of patients treated with 40 mg. A minor increase in shift from none or trace to + has been reported with the 20 mg dose. In most cases, proteinuria decreases or disappears spontaneously on continued therapy. Review of data from reported clinical trials and post-marketing experience to date has not identified a causal association between proteinuria and acute or progressive renal disease.

Haematuria has been reported in patients treated with rosuvastatin and the occurrence has been reported to be low.

Skeletal muscle effects: Effects on skeletal muscle e.g. myalgia, myopathy (including myositis) and, rarely, rhabdomyolysis with and without acute renal failure have been reported in rosuvastatin-treated patients with all doses and in particular with doses $>$ 20 mg.

A dose-related increase in CK levels has been reported in patients taking rosuvastatin; the majority of cases were mild, asymptomatic and transient. If CK levels are elevated ($>$ 5 x ULN), treatment should be discontinued.

Liver effects: As with other HMG-CoA reductase inhibitors, a dose-related increase in transaminases has been reported in a small number of patients taking rosuvastatin; the majority of cases were mild, asymptomatic and transient.

The following adverse events have been reported with some statins:

Sexual dysfunction.

Exceptional cases of interstitial lung disease, especially with long term therapy.

Tendon disorders, sometimes complicated by rupture.

The reporting rates for rhabdomyolysis, serious renal events and serious hepatic events (consisting mainly of increased hepatic transaminases) is higher at the 40 mg dose.

Pediatric patients 10 to 17 years of age: In boys and postmenarchal girls, the safety and tolerability profile of rosuvastatin 5 to 20 mg daily has been reported to be generally similar to that of placebo.

However, elevations in serum creatine phosphokinase (CK) $>10 \times$ ULN has been reported more frequently in rosuvastatin compared with placebo-treated children.

OVERDOSE

There is no specific treatment in the event of overdose. In the event of overdose, the patient should be treated symptomatically and supportive measures instituted as required. Liver function and CK levels should be monitored. Hemodialysis is unlikely to be of benefit.

PHARMACODYNAMIC AND PHARMACOKINETIC PROPERTIES

Pharmacodynamic properties

Rosuvastatin is a selective and competitive inhibitor of HMG-CoA reductase, the rate-limiting enzyme that converts 3-hydroxy-3-methylglutaryl coenzyme A to mevalonate, a precursor for cholesterol. The primary site of action of rosuvastatin is the liver, the target organ for cholesterol lowering.

Rosuvastatin increases the number of hepatic LDL receptors on the cell-surface, enhancing uptake and catabolism of LDL and it inhibits the hepatic synthesis of VLDL, thereby reducing the total number of VLDL and LDL particles.

Rosuvastatin reduces elevated LDL-cholesterol, total cholesterol and triglycerides and increases HDL-cholesterol. It also lowers ApoB, nonHDL-C, VLDL-C, VLDL-TG and increases ApoA-I. Rosuvastatin also lowers the LDL-C/HDL-C, total C/HDL-C and nonHDL-C/HDL-C and the ApoB/ApoA-I ratios.

A therapeutic effect is obtained within 1 week following treatment initiation and 90% of maximum response is achieved in 2 weeks. The maximum response is usually achieved by 4 weeks and is maintained after that.

Pharmacokinetic properties

Absorption

The peak plasma concentrations of rosuvastatin are reached 3 to 5 hours following oral dosing. Both peak concentration (C_{max}) and area under the plasma concentration time curve (AUC) increased in approximate proportion to rosuvastatin dose. The absolute bioavailability of rosuvastatin is approximately 20%.

Administration of rosuvastatin with food decreased the rate of drug absorption by 20% as assessed by C_{max} , but there was no effect on the extent of absorption as assessed by AUC.

Plasma concentrations of rosuvastatin do not differ following evening or morning drug administration.

Significant LDL-C reductions have been reported when rosuvastatin is given with or without food and regardless of the time of day of drug administration.

Distribution

Mean volume of distribution at steady state of rosuvastatin is approximately 134 litres. Rosuvastatin is 88 % bound to plasma proteins, mostly albumin. This binding is reversible and independent of plasma concentrations.

Metabolism

Rosuvastatin is not extensively metabolized; approximately 10% of dose is recovered as metabolite. The major metabolite is N-desmethyl rosuvastatin, which is formed principally by cytochrome P450 2C9, and has approximately one-sixth to one-half the HMG-CoA reductase inhibitory activity of rosuvastatin. Overall, greater than 90% of active plasma HMG-CoA reductase inhibitor activity is accounted by rosuvastatin.

Excretion

Following oral administration, rosuvastatin and its metabolite are primarily excreted in the faeces (90%). The elimination half-life ($t_{1/2}$) of rosuvastatin is approximately 19 hours.

After an intravenous dose, approximately 28% of total body clearance was via the renal route and 72% by the hepatic route.

Special populations

Age and sex

There was no clinically relevant effect of age or sex on the pharmacokinetics of rosuvastatin.

Race

An approximate 2-fold elevation in median AUC has been reported in Asian subjects compared with Caucasians. No clinically relevant differences in pharmacokinetics have been reported among Caucasian, Hispanic and Black or Afro-Caribbean groups.

STORAGE

Store below 30°C, protected from moisture.

KEEP ALL MEDICINES OUT OF REACH OF CHILDREN

SUPPLY

Blister strips of 4 x 7's; Blister strips of 10 x 10's and Blister strips of 3 x 10's

Date of Revision: May 2023

Product Registration Holder / Manufactured by:

RANBAXY (MALAYSIA) SDN BHD

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